An encrypted traffic classification method based on contrastive learning

siyuan tian
Beijing University of Posts and Telecommunications

Yating Gao
State Grid Information & Telecommunication Branch, Beijing 100761, China

Guoquan Yuan
Information & Telecommunication Branch, State Grid Jiangsu Electric Power Co. Ltd, Nanjing, China

Ru ZHANG*
Beijing University of Posts and Telecommunications

Jinmeng Zhao
State Grid Information & Telecommunication Branch, Beijing 100761, China

Song Zhang
Information & Telecommunication Branch, State Grid Jiangsu Electric Power Co. Ltd, Nanjing, China

ABSTRACT

Network traffic classification has become an important part of network management, which is conducive to realizing intelligent network operation and maintenance, improving network quality of service (QoS), and ensuring network security. With the rapid development of various applications and protocols, more and more encrypted traffic appears in the network. Due to the loss of semantic information after traffic encryption, poor content intelligibility, and difficulty in feature extraction, traditional detection methods are no longer applicable. Existing solutions mainly rely on the powerful feature self-learning ability of end-to-end deep neural networks to identify encrypted traffic. However, such methods are overly dependent on data size, and it has been experimentally proven that it is often difficult to achieve satisfactory results when validating across datasets. In order to solve this problem, this paper proposes an encrypted traffic identification method based on contrastive learning. First, the clustering method is used to expand the labeled data set. When the encrypted traffic features are difficult to extract, it is only necessary to learn the feature space to achieve discrimination.more suitable for encrypted traffic identification. When validating across datasets, only fine-tuning is required on a small amount of labeled data to achieve good recognition results. Compared with the end-to-end learning method, there is an improvement of about 5%.

CCS CONCEPTS •Security and privacy •Network security •Security protocols

KEYWORDS

encrypted traffic, contrastive learning, feature extraction

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*Corresponding Author email: zhangru@bupt.edu.cn

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1 INTRODUCTION

Traffic classification is an essential task for traffic engineering, network management, quality of service (QoS), and network security. Its purpose is to identify categories of traffic from various applications or web services. In recent years, with the popularization of traffic encryption technology and the rapid growth of network throughput, traffic encryption has been widely used on the Internet. Numerous services and applications use encryption algorithms to protect their information. It is becoming increasingly difficult to accurately identify encrypted traffic in real time. The classification and identification framework of encrypted traffic based on deep learning meets the needs of end-to-end classification and identification. Using the original traffic content as the model input, 1D-CNN, 2D-CNN, etc. are used for feature extraction. Although validating on a single dataset works well, it is often difficult to achieve ideal results when validating across datasets. Therefore, how to capture the implicit and robust patterns in various encrypted traffic and support accurate and general traffic classification is the key to achieve high network security and effective network management.

Contrastive learning has demonstrated its advantages for feature extraction in a wide range of applications across domains, such as visual language and computer vision. Aiming at the above problems, this paper designs a method for identifying encrypted traffic based on contrastive learning. After the encrypted traffic data is preprocessed, in the case of only a small number of labeled samples, first use the clustering method to expand the data volume of the training set, and obtain a pre-trained model based on supervised comparative learning training, and then use a small number of samples for fine-tuning for downstream use. Task. The experimental results show that the generalization ability of this method is improved compared with the end-to-end encrypted traffic identification method.

2 RELATED WORK

2.1 Identification of Encrypted Traffic

2.1.1 Payload-based approach. Bonfiglio et al. [1] utilize the unencrypted 4-byte header and the first 4 of the payload based on the Pearson chi-square test and the Naive Bayes classifier. Chi-square detection is performed on each byte to extract the fingerprint of the protocol frame structure for Skype traffic identification. Bai Yu [2] established a first-order homogeneous Markov chain by using the secure socket protocol (SSL) and transport layer protocol (TLS) message types as eigenvalues, and calculated the initial state and end state by using the frequency statistics of state appearance and transition to build a Markov model (fingerprint model) of the data.

2.1.2 Statistical feature-based Methods. According to the unencrypted bit stream data of different lengths in the link layer, Xing Meng [3] et al. used four classical methods such as symbol frequency detection in NIST detection to extract and identify its features.

2.1.3 Machine-learning Based Methods. Sun et al. [4] proposed a hybrid approach to classify encrypted traffic. In this approach, the SSL/TLS protocol is first identified using a signature matching method. A Naive Bayesian machine learning algorithm is then applied to identify encrypted applications under the SSL/TLS protocol. Alshammari [5] studied the performance of C5.0, AdaBoost, and Genetic Programming (GP) algorithms for recognizing Voice over IP (VoIP) traffic. Finally, it is concluded that applying suitable sampling and machine learning methods is important for VoIP traffic classification. In addition, other feature-based methods are widely used in traffic classification[6, 7, 8].

2.1.4 Deep-learning Based Methods. Vinh et al. used neural networks to classify news and proved the effectiveness of neural networks in sequence classification[9].Wang et al. [10] first proposed a classification and recognition framework based on one-dimensional CNN, which overcomes the problem that traditional machine learning algorithms are difficult to obtain the global optimal value, and meets the needs of end-to-end classification and recognition. Based on this, the subsequent research designs different preprocessing methods of encrypted traffic data using two-dimensional CNN or one-dimensional CNN to design the recognition framework [11, 12].

2.2 Contrastive Learning

This simple idea has been widely used in self-supervised learning of image features and pre-training of model parameters in recent years. Chen uses two augmentation mechanisms to generate two interrelated views for each example, making related views attract each other while repelling other examples. Representations are learned by maximizing the consistency between different augmented views of the same data example through a contrastive loss in the latent space [13]. Based on the given label information, Khosla et al. make the samples with the same label as positive samples and other samples as negative samples. Using the supervised contrastive learning method has achieved better results on the ImageNet dataset than the traditional model based on the cross-entropy loss function [14].

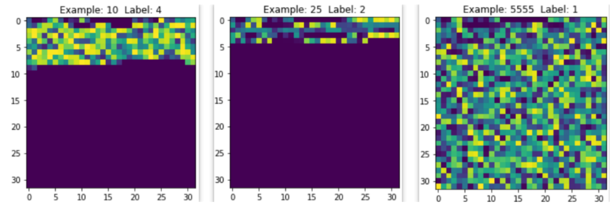

Figure 1: Visualization of traffic preprocessing results

3 ALGORITHM DESIGN

3.1 Granularity Selection

The split granularity of network traffic includes TCP connections, flows, sessions, services and hosts. A session is a collection of flows, and a flow is defined as all packets with the same five-tuple. The five-tuple contains source IP, destination IP, source port, Destination port and transport level protocols, sessions are bidirectional flows, covering both flows where source IP/port and destination IP/port are interchangeable. To construct the set of sessions, we describe all packets in the original network traffic as a group $P = \{p^1, p^2, \cdot s, p^{|p|}\}$, and each packet in the group is represented as $p^i = (x^i, b^i, t^i)$, $i = 1, 2, 3, \cdot s$. x^i represents a quintuple, b^i represents the byte size of the corresponding packet, and t^i represents the time when the packet starts to be sent. A group P in the original network traffic is divided into multiple subsets, each subset represents a flow, and a subset is described in the form of $f = (x, b, d, t)$. x represents the same quintuple, b represents the sum of the sizes of all packets in a stream, d_t represents the duration of the stream, and t represents the time when the first packet of the stream starts outputting. During a period of time, multiple flows with the same or opposite quintuple information form a session S, and the original network traffic can be converted into a set of multiple sessions, that is, the flow $T = (S_1, S_2, S_3, \cdot s)$.

3.2 Unlabeled data processing

Supervised contrastive learning relies on deep features, which depend on data size. In order to expand the amount of labeled data in the dataset, the PCA method is first used to reduce the dimension of high-dimensional traffic data to obtain a low-dimensional representation of the data, so as to avoid the problem of dimension disaster caused by too high feature dimensions. Then, the labeled data and unlabeled data are clustered together by clustering method, and the unlabeled data with higher confidence is determined according to the proportion of labeled data in different clusters to the data in the whole cluster, and pseudo-labels are added to expand the labeled data set.

3.3 Encrypted traffic identification method based on supervised contrastive learning

This paper proposes an encrypted traffic identification method based on supervised contrastive learning. For the first time, the supervised contrastive learning method is used in the encrypted traffic identification pre-training model. The pre-training model consists of two neural networks, one of which is used for encrypted traffic feature extraction called encoding. The projector E, and the

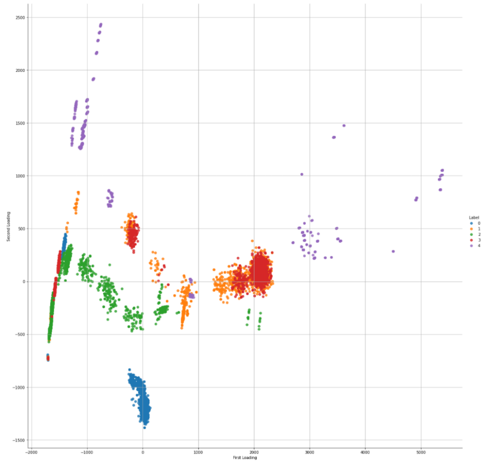

Figure 2: Dimensionality reduction visualization of encrypted traffic features

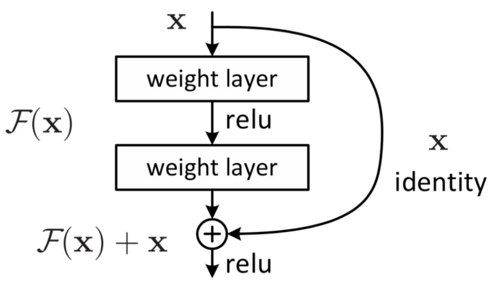

Figure 3: Internal composition of residual structure

other for converting high-dimensional features to low-dimensional features is called projector P. In terms of model selection, this paper selects the resnet model proposed by He Yuming et al. of Microsoft Research in 2015 as the encoder architecture.

Suppose the input is x, and the mapping learned by two fully connected layers is $H(x)$. Assuming that the dimensions of $H(x)$ and x are the same, then fitting $H(x)$ is equivalent to fitting residual function $H(x)-x$. Let residual function $F(x) = H(x)-x$, then The original function becomes $F(x) - x$, so a cross-layer connection is added directly on the basis of the original network. In essence, the objective function $H(x)$ is not changed, and the network structure is split into two branches, one of which is the residual. To map $F(x)$, one branch is the identity map x, so the network only needs to learn the residual map $F(x)$.Considering the small size of the dataset, this paper only uses the resnet18 model with the fewest parameters for training.

4 EXPERIMENT

4.1 Dataset Selection

We validate our method using two public encrypted traffic datasets, ISCX VPN-nonVPN (Draper-Gil et al., 2016)[15] and USTC-TFC2016[16]. The ISCX VPN-nonVPN traffic dataset used for evaluation includes 7 regular encrypted traffic and 7 protocol encapsulated traffic. Consists of 15 apps such as Facebook, Youtube, Netflix. Selected applications are encrypted using various security protocols, including HTTPS, SSL, SSH, and proprietary protocols. The USTC-TFC2016 dataset contains encrypted traffic of 10 application types such as MySQL, Gmail, and Outlook. Both datasets are collected from real network environments.

Table 1: ISCX VPN-nonVPN classification results

Traffic type	Content
Chat	Aim、Facebook、Hangouts、Icq、Skype
E-mail	Email
File	Ftps、Skype
Streaming	Spotify、Vimeo、Youtube
Voip	Facebook、Hangouts、Voipbuster

Table 2: USTC-TFC2016 classification results

Traffic type	Content
Chat	Skype
E-mail	GEmail、Outlook
File	Ftp、SMB
P2p	BitTorrent
Voip	Facetime

We manually label the two datasets, ISCX VPN-nonVPN and USTC-TFC2016, respectively, so that the two datasets are divided into five categories. Table 1 and Table 2 show our classification results.

4.2 Experimental results and analysis

In the TFC-2016 dataset, based on 9000 pieces of labeled data, we first use PCA to reduce the dimension of the data and select five cluster centers for clustering using K-Means, adding pseudo-labels to the unlabeled data, as Supervised contrastive learning prepares data. PCA technology can achieve dimensionality reduction to simplify the model, while maintaining the information of the original data to the greatest extent.

After clustering and adding pseudo-labels, 14,000 pieces of data with pseudo-labels are obtained. In order to verify the generalization ability of the model, we will use 23,000 pieces of data to use end-to-end and supervised comparative learning methods to train the feature extraction model. After the training is completed, a fully connected layer is added, and a small amount of data is used for fine-tuning in the ISCX dataset to finally verify the model's encrypted traffic recognition effect across datasets.

In order to verify that the generalization ability of the pre-trained model after adding pseudo-labeled data is improved compared with before, this paper sets up a control experiment. In addition to the above set of experiments, only the labeled data is used for model training, and cross-dataset verification is still performed.

The above experimental results are organized as shown in the following table.

Table 3: TFC dataset clustering results

cluster id	Label_-1	Label0	Label1	Label2	Label3	Label4
0	10812	109	190	1649	776	0
1	9344	0	1806	159	1470	316
2	3525	0	7	0	0	1585
3	2933	405	0	0	0	0
4	3905	266	85	260	196	40

Table 4: Summary of experimental results (accuracy rate)

Amount of training data	methods	
	end-to-end learning	Supervised Contrastive Learning
9000	79	86
23000	84.5	90.55

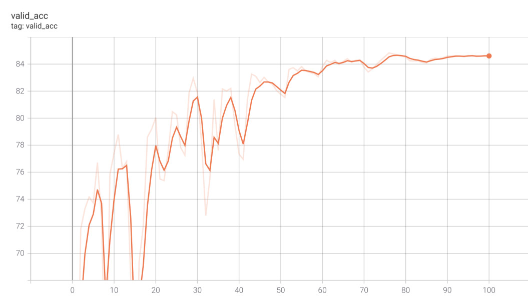

Figure 4: End-to-end training cross-dataset validation results

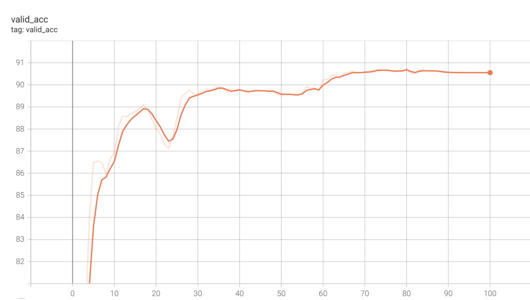

Figure 5: Contrastive learning cross-dataset validation results

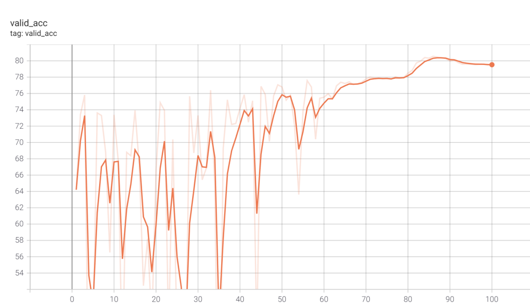

Figure 6: End-to-end training cross-dataset validation results

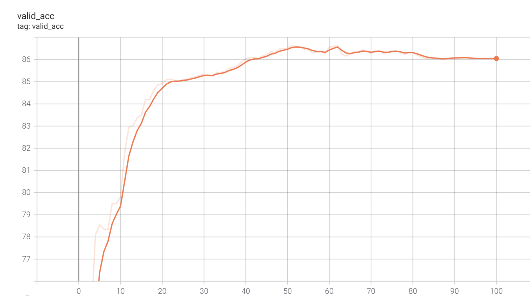

Figure 7: Contrastive learning cross-dataset validation results

It can be seen from the above results that when the model converges, the model trained by comparative learning has better generalization ability than the end-to-end model, and can achieve better results in cross-dataset validation, and under the same conditions. In this case, the pre-training model based on contrastive learning can achieve the effect of end-to-end training in only 10 training rounds. In addition, under the same other conditions, in order to verify the validity of the pseudo-label data, adding pseudo-label data during the training process slightly improves the effect of the two training modes.

5 CONCLUSION

After the traffic is encrypted, the intelligibility of the payload content is poor, and feature extraction is difficult. The traditional end-to-end model is difficult to understand the true meaning of the payload content. Comparative learning only needs to learn the distinction in the feature space, without the need for a deep understanding of the encrypted content. It is suitable for feature extraction of encrypted traffic and can get better results in downstream tasks after fine-tuning. In addition, in order to solve the problem that a large amount of labeled data is needed for supervised comparative learning, this paper adopts the clustering method to select data with high confidence to expand the labeled data set. The experimental results show that after adding pseudo-labeled data, the generalization of the model The ability is slightly improved compared to the previous one, which shows the effectiveness of this method for feature extraction of encrypted traffic.

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